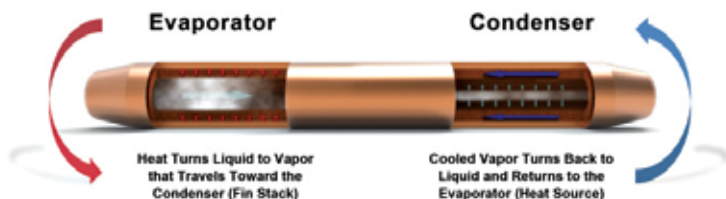


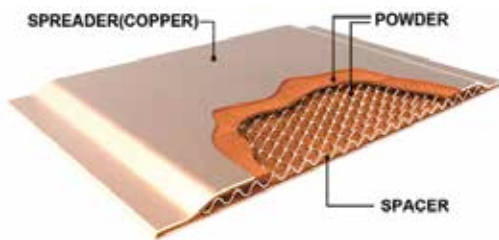
is slightly reduced. A better approach is to have a thin but flat copper base plate which has the heat pipes running through it. And the heat pipes are very close to the base plate. The base plate can even be tin plated for corrosion prevention. We'd like to point out that the temperature difference here is absolutely minimal as well.



A heat pipe

Heat pipes and vapour chambers

Then there are the heat pipes and the vapour chambers. Fundamentally, the two operate in the exact same way. Let's take the example of the heatpipe. It's simply a hollow cylindrical structure that's usually made out of copper or any other metal with high thermal conductivity. It's closed on all ends and has a coolant on the inside. When the part of the heatpipe which is contact with the heat source heats up, the coolant on the inside boils and turns into vapour. This vapour then travels to the cooler part of the heat pipe and condenses back into liquid form. This phase change happens because heat was transferred from the coolant to the fins. This is called a two-phase cooling



A vapour chamber

system. The insides of heatpipes have fine grooves running along the length. The coolant which has changed from vapour state to liquid state at the cooler side, travels back to the hot end through these grooves

because of capillary action. So if you pay close attention to the diagram, then the vapour has a dedicated path along the middle to move from the hot to the cold zone and the liquid coolant has a dedicated path along the grooves on the sides to return to the hot zone.